Convolution

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$x[n] = \{-1, 0, 1, 2, 3\}$
$\uparrow$

$h[n] = \{-2, -1, 0\}$
$\uparrow$

length, $N = N_1 + N_2 - 1 \Rightarrow N = 5 + 3 - 1 = 7$

<u>range</u>

$k_L = x_L = 0$ $\qquad 0 \leq k \leq 4$

$k_h = x_h = 4$

$n_L = x_L + h_L = 0 + 0 = 0$ $\qquad 0 \leq n \leq 6$

$n_h = x_h + h_h = 4 + 2 = 6$

$$y(n) = \sum_{0}^{4} x(k)\, h(n-k)$$

$n = 0, \quad y(0) = \sum_{0}^{4} x(k)\, h(-k)$

$$= x(0)h(0) + x(1)h(-1) + x(2)h(-2) + x(3)h(-3) + x(4)h(-4)$$

$$= (-1)(-2) + (0)(0) + (1)(0) + (2)(0) + (3)(0)$$

$$= 2$$

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$n=1,\ y(1) = \sum_{0}^{4} x(k)\, h(1-k)$

$= x(0)h(1) + x(1)h(0) + x(2)h(-1) + x(3)h(-2) + x(4)h(-3)$

$= (-1)(-1) + 0 + (1)(0) + 0 + 0$

$= 1$

$n=2,\ y(2) = \sum_{0}^{4} x(k)\, h(2-k)$

$= x(0)h(2) + x(1)h(1) + x(2)h(0) + x(3)h(-1) + x(4)h(-2)$

$= (-1)(0) + 0 + (1)(-2) + (2)(0) + 0$

$= -2$

$n=3,\ y(3) = \sum_{0}^{4} x(k)\, h(3-k)$

$= x(0)h(3) + x(1)h(2) + x(2)h(1) + x(3)h(0) + x(4)h(-1)$

$= (-1)(0) + 0 + (1)(-1) + (2)(-2) + (3)(0)$

$= -1 - 4$

$= -5$

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$n=4,\quad y(4) = \sum_{0}^{4} x(k)h(4-k)$

$= x(0)h(4) + x(1)h(3) + x(2)h(2) + x(3)h(1) + x(4)h(0)$

$= (-1)(0) + 0 + (1)(0) + (2)(-1) + (3)(-2)$

$= -2-6$

$= -8$

$n=5,\quad y(5) = \sum_{0}^{4} x(k)h(5-k)$

$= x(0)h(5) + x(1)h(4) + x(2)h(3) + x(3)h(2) + x(4)h(1)$ $\cancel{+x(4)h(1(}$

$= (-1)(0) + 0 + (1)(0) + (2)(0) + (3)(-1)$

$= -3$

$n=6,\quad y(6) = \sum_{0}^{4} x(k)h(6-k)$

$= x(0)h(6) + x(1)h(5) + x(2)h(4) + x(3)h(3) + x(4)h(2)$

$= (-1)(0) + 0 + (1)(0) + (2)(0) + (3)(0)$

$= 0$

$y[n] = \{\underset{\uparrow}{2}, 1, -2, -5, -8, -3, 0\}$

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